

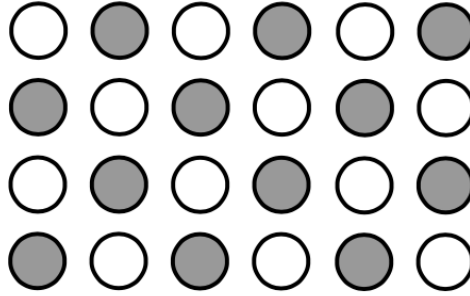


**The CENTRE for EDUCATION
in MATHEMATICS and COMPUTING**

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**2025 Gauss Grade 8
Solutions**

1. In the diagram, how many of the 24 circles are shaded?



- (A) 10 (B) 12 (C) 14 (D) 16 (E) 18

Source: 2025 Gauss Grade 8 #1

Primary Topics: Counting and Probability

Secondary Topics: Counting | Patterning/Sequences/Series

Answer: B

Solution:

In each row, exactly 3 of the 6 or one-half of the circles are shaded. Thus, one-half of the 24, that is, 12 circles are shaded. Alternately, we could just count the 12 shaded circles.

2. Seong-hun had 36 dried apricot pieces that he gave to his 4 children. Each child received the same number of pieces. How many pieces did each child receive?

- (A) 12 (B) 6 (C) 3 (D) 9 (E) 10

Source: 2025 Gauss Grade 8 #2

Primary Topics: Number Sense

Secondary Topics: Counting | Fractions/Ratios | Operations

Answer: D

Solution:

If each of 4 children shared 36 pieces equally, then each child received $\frac{36}{4} = 9$ pieces.

3. Recycling is picked up every two weeks. Recycling was last picked up on May 12. On what date is the recycling picked up next?
(A) May 19 (B) May 20 (C) May 25 (D) May 26 (E) May 27

Source: 2025 Gauss Grade 8 #3

Primary Topics: Counting and Probability

Secondary Topics: Counting | Operations

Answer: D

Solution:

There are 7 days in each week, and so there are $2 \times 7 = 14$ days in two weeks.

Thus, the date two weeks from May 12 is May 26 (since $12 + 14 = 26$).

4. At 8 : 45a. m., Aisha starts a movie that is 2 hours and 45 minutes long. If she watches the movie all the way through without a pause or a break, at what time will it finish?
(A) 10 : 30a. m. (B) 11 : 15a. m. (C) 11 : 30a. m. (D) 10 : 50a. m.
(E) 10 : 45a. m.

Source: 2025 Gauss Grade 8 #4

Primary Topics: Counting and Probability | Number Sense

Secondary Topics: Counting | Digits | Operations

Answer: C

Solution:

Solution 1:

The time 2 hours after 8:45 a.m. is 10:45 a.m., and the time 45 minutes after 10:45 a.m. is 11:30 a.m.

Solution 2:

Two hours and 45 minutes is 15 minutes less than 3 hours.

The time 3 hours after 8:45 a.m. is 11:45 a.m., and the time 15 minutes before 11:45 a.m. is 11:30 a.m.

5. If $7x - 3 = 60$, the value of x is

- (A) 9 (B) 7 (C) 10 (D) 6 (E) 8

Source: 2025 Gauss Grade 8 #5

Primary Topics: Algebra and Equations

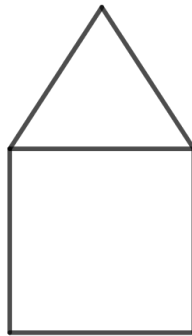
Secondary Topics: Equations Solving

Answer: A

Solution:

If $7x - 3 = 60$, then $7x = 60 + 3$, so $7x = 63$ and $x = \frac{63}{7} = 9$.

6. A figure is made by placing a triangle on top of a square, as shown.



The triangle is coloured either red or yellow. The square is coloured either blue or purple or green. How many different ways can the figure be coloured?

- (A) 5 (B) 3 (C) 9 (D) 6 (E) 2

Source: 2025 Gauss Grade 8 #6

Primary Topics: Counting and Probability

Secondary Topics: Counting | Operations | Polygons

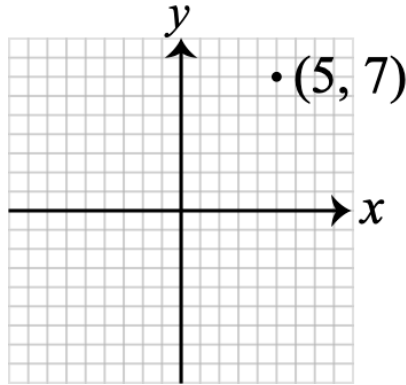
Answer: D

Solution:

There are 2 ways to colour the triangle and 3 ways to colour the square, and so there are $2 \times 3 = 6$ ways to colour the figure.

We can write the 6 colourings of the figure as ordered pairs in which the first colour listed in the pair is the colour of the triangle and the second is the colour of the square. These are: (red, blue), (red, purple), (red, green), (yellow, blue), (yellow, purple), and (yellow, green).

7. The point $(5, 7)$ is plotted on the graph shown.



When $(5, 7)$ is reflected in the x -axis, the resulting point is

- (A) $(5, -7)$ (B) $(-5, -7)$ (C) $(5, 7)$ (D) $(7, 5)$ (E) $(-7, -5)$

Source: 2025 Gauss Grade 8 #7

Primary Topics: Geometry and Measurement

Secondary Topics: Transformations | Coordinate Geometry

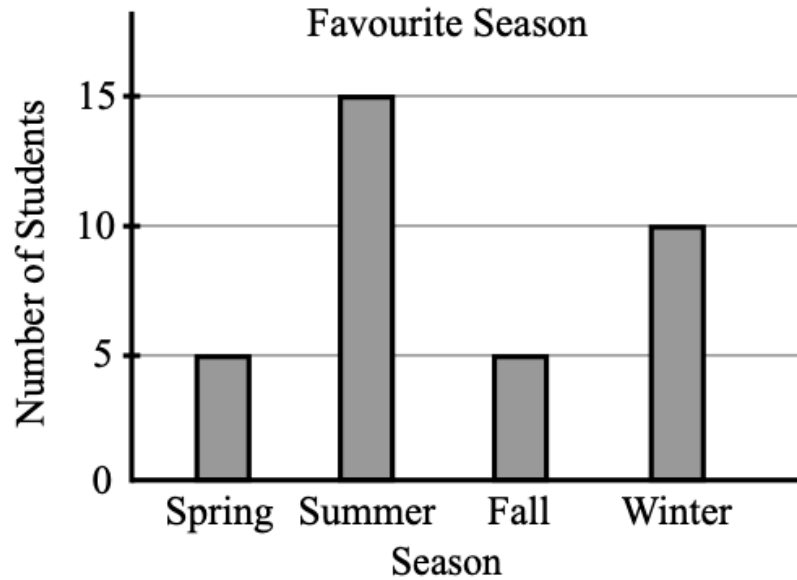
Answer: A

Solution:

When the point $(5, 7)$ is reflected in the x -axis, the resulting point is $(5, -7)$.

When a point is reflected in the x -axis, its x -coordinate does not change and its y -coordinate changes sign (provided that the original point is not on the x -axis).

8. Each of Mrs. Myer's students voted exactly once for their favourite season.



Which of the following statements about the results in the graph shown is *false*?

- (A) Five students voted for Fall.
- (B) Winter received more votes than Spring.
- (C) Thirty-five students participated in this survey.
- (D) More than half of the students voted for Summer.
- (E) Fall and Spring received the same number of votes.

Source: 2025 Gauss Grade 8 #8

Primary Topics: Data Analysis

Secondary Topics: Graphs | Counting

Answer: D

Solution:

Reading from the graph, 5 students voted for Spring, 15 voted for Summer, 5 voted for Fall, and 10 voted for Winter.

Therefore, (A) Fall and Spring received the same number of votes, (B) Winter received more votes than Spring, (C) 35 students participated in the survey, and (E) 5 students voted for Fall, are all true statements.

Since 15 students voted for Summer and 15 is less than half of 35, then (D) is the statement that is false.

9. Ruhab wrote the list 5, 2, 8, 7, 9 and then erased one of the five digits. The sum of the remaining four digits was a multiple of 4. Which number did she erase?

- (A) 5 (B) 2 (C) 8 (D) 7 (E) 9

Source: 2025 Gauss Grade 8 #9

Primary Topics: Number Sense

Secondary Topics: Divisibility | Counting | Operations

Answer: D

Solution:

Solution 1:

The sum of the original five digits is $5 + 2 + 8 + 7 + 9 = 31$.

The largest multiple of 4 that is less than 31 is 28, and 28 is 3 less than 31.

However, 3 is not among the list of five digits, and thus Ruhab cannot erase a 3.

The next largest multiple of 4 that is less than 31 is 24.

Since $31 - 24 = 7$, and 7 does appear in the original list of five digits, then if Ruhab erases the 7, the sum of the remaining four digits is a multiple of 4.

(We may confirm that $5 + 2 + 8 + 9 = 24$.)

Solution 2:

We may erase each of the five digits one at a time and in each case, determine the sum of the remaining four digits.

Doing so, we get

$$5 + 2 + 8 + 7 = 22; 5 + 2 + 8 + 9 = 24; 5 + 2 + 7 + 9 = 23;$$

$$5 + 8 + 7 + 9 = 29; 2 + 8 + 7 + 9 = 26$$

Of these sums, only 24 is a multiple of 4 and so Ruhab erased the digit 7.

10. An integer from 3 to 20, inclusive, is randomly selected. What is the probability that the integer selected is a perfect square?

- (A) $\frac{3}{20}$ (B) $\frac{1}{9}$ (C) $\frac{1}{6}$ (D) $\frac{1}{10}$ (E) $\frac{2}{9}$

Source: 2025 Gauss Grade 8 #10

Primary Topics: Counting and Probability

Secondary Topics: Probability | Counting | Fractions/Ratios

Answer: 10

Solution:

The perfect squares are $1^2 = 1$, $2^2 = 4$, $3^2 = 9$, and so on.

There are 18 integers between 3 and 20 inclusive.

Of these 18 integers, only $2^2 = 4$, $3^2 = 9$ and $4^2 = 16$ are perfect squares.

(We note that $1^2 = 1$ and $5^2 = 25$ are not among the integers given.)

The probability of randomly selecting a perfect square from the integers given is $\frac{3}{18} = \frac{1}{6}$.

11. What number goes in the box so that $\frac{28}{32} + \frac{1}{\square} = 1$?
- (A) 24 (B) -3 (C) 7 (D) 16 (E) 8

Source: 2025 Gauss Grade 8 #11

Primary Topics: Algebra and Equations

Secondary Topics: Equations Solving | Fractions/Ratios | Operations

Answer: E

Solution:

Since $\frac{28}{32} + \frac{1}{\square} = 1$ and $\frac{28}{32} + \frac{4}{32} = \frac{32}{32} = 1$, then $\frac{1}{\square} = \frac{4}{32}$.

Reducing $\frac{4}{32}$ to lowest terms, we get $\frac{4}{32} = \frac{1}{8}$, and so the number that goes in the box is 8.

12. Leticia can walk 1.5km in 20minutes. Walking at this same rate, how far does Leticia walk in 4hours?
- (A) 18km (B) 30km (C) 22.5km (D) 15km (E) 4.5km

Source: 2025 Gauss Grade 8 #12

Primary Topics: Algebra and Equations

Secondary Topics: Rates | Fractions/Ratios | Decimals

Answer: A

Solution:

There are 60 minutes in one hour.

Since $3 \times 20 \text{ minutes} = 60 \text{ minutes}$, and $3 \times 1.5 \text{ km} = 4.5 \text{ km}$, then Leticia can walk 4.5 km in one hour.

Walking at this same rate, Leticia can walk $4 \times 4.5 \text{ km} = 18 \text{ km}$ in 4 hours.

13. A list of one-digit integers contains exactly one 1, two 2s, three 3s, four 4s, five 5s, and six 6s. What is the median of this list?

- (A) 2 (B) 3 (C) 4 (D) 5 (E) 6

Source: 2025 Gauss Grade 8 #13

Primary Topics: Data Analysis | Number Sense

Secondary Topics: Averages | Counting

Answer: D

Solution:

Ordered from smallest to largest, the list is

1, 2, 2, 3, 3, 3, 4, 4, 4, 4, 5, 5, 5, 5, 5, 6, 6, 6, 6, 6, 6

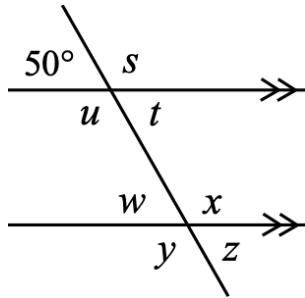
This list contains $1 + 2 + 3 + 4 + 5 + 6 = 21$ integers.

The median is the middle number in an ascending (or descending) list of numbers.

The 11th integer in the list of 21 integers above has 10 integers before it and 10 integers following it, and thus is the median of the list.

The 11th integer and the median of the given list is 5.

14. In the diagram, which of the following pairs of angles have measures whose sum is equal to 180° ?



- (A) w and z (B) x and y (C) u and x (D) t and y (E) s and x

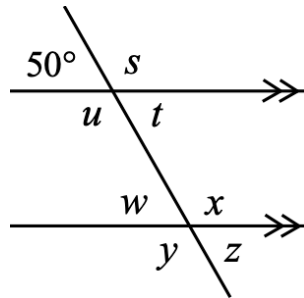
Source: 2025 Gauss Grade 8 #14

Primary Topics: Geometry and Measurement

Secondary Topics: Equations Solving | Angles | Measurement

Answer: D

Solution:



Opposite angles are equal in measure and so $t = 50^\circ$ since it is opposite the angle whose measure is given as 50° .

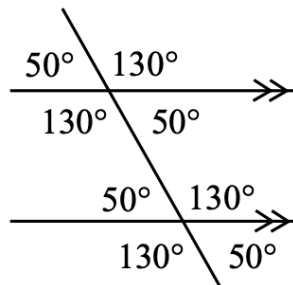
By a property of parallel lines, the angles whose measures are t and x form a "C pattern", and are supplementary.

Thus $t + x = 180^\circ$ or $x = 180^\circ - 50^\circ = 130^\circ$.

Since the angle whose measure is y is opposite the angle whose measure is x , then $y = x = 130^\circ$.

Since $t + y = 50^\circ + 130^\circ = 180^\circ$, then from the given answers, t and y are the pair of angles whose measures sum to 180° .

Of the answers given, we confirm that no other pair of angles have measures whose sum is 180° by determining the measures of each of the missing angles, as shown.



15. The ages of three students are consecutive integers. Their mean (average) age is 13. A fourth student joins the group and the mean of their four ages is 14. How old is the fourth student?
- (A) 15 (B) 18 (C) 16 (D) 14 (E) 17

Source: 2025 Gauss Grade 8 #15

Primary Topics: Number Sense | Algebra and Equations

Secondary Topics: Averages | Equations Solving | Counting

Answer: E

Solution:

Solution 1:

The mean age of the first three students is 13, and so the sum of their ages is $13 \times 3 = 39$.

The mean age of the four students is 14, and so the sum of their ages is $14 \times 4 = 56$.

The age of the fourth student is the difference between these two sums, which is $56 - 39 = 17$ years old.

Solution 2:

In an ordered list of 3 consecutive integers, the average of the list is always the middle integer. Can you see why?

Therefore, three students whose ages are consecutive integers and whose mean age is 13 are 12, 13 and 14 years old.

Suppose the fourth student is x years of age.

Since the mean age of the four students is 14, then $\frac{12 + 13 + 14 + x}{4} = 14$ or $39 + x = 14 \times 4$, and so $x = 56 - 39 = 17$.

The fourth student is 17 years old.

16. At Doggy Daycare, there is one dog for every bowl of food, two dogs for every bowl of water, and three dogs for every bowl of treats. Every dog gets a serving of food, water and treats. If there are a total of 77 bowls, how many dogs are there?
- (A) 35 (B) 77 (C) 42 (D) 11 (E) 24

Source: 2025 Gauss Grade 8 #16

Primary Topics: Algebra and Equations

Secondary Topics: Equations Solving | Counting | Expressions

Answer: C

Solution:

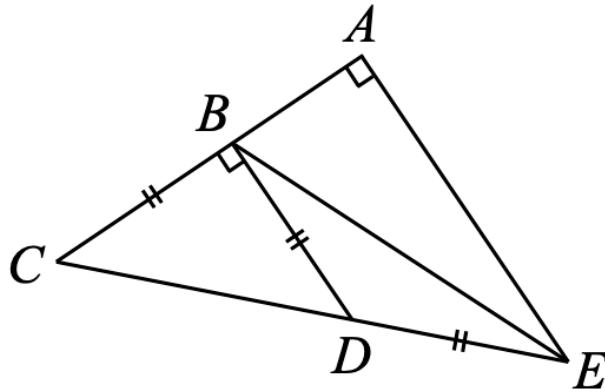
There is 1 dog for every bowl of food, and so if there were 6 dogs, then there would be 6 bowls of food.

There are 2 dogs for every bowl of water, and so these same 6 dogs would need $\frac{6}{2} = 3$ bowls of water.

There are 3 dogs for every bowl of treats, and so the 6 dogs would need $\frac{6}{3} = 2$ bowls of treats. Thus, 6 dogs require 6 bowls of food, 3 bowls of water, and 2 bowls of treats, or $6 + 3 + 2 = 11$ bowls in total.

There are 11 bowls for every 6 dogs, and since there are $77 = 11 \times 7$ bowls, then there are 7 groups of 6 dogs, or $6 \times 7 = 42$ dogs.

17. Points B and D lie on sides AC and CE , respectively, of $\triangle ACE$, as shown.



If $\angle CAE = \angle CBD = 90^\circ$ and $CB = BD = DE$, the measure of $\angle ABE$ is
 (A) 60° (B) 67.5° (C) 70° (D) 75° (E) 52.5°

Source: 2025 Gauss Grade 8 #17

Primary Topics: Geometry and Measurement

Secondary Topics: Angles | Expressions | Measurement

Answer: B

Solution:

There is 1 dog for every bowl of food, and so if there were 6 dogs, then there would be 6 bowls of food.

There are 2 dogs for every bowl of water, and so these same 6 dogs would need $\frac{6}{2} = 3$ bowls of water.

There are 3 dogs for every bowl of treats, and so the 6 dogs would need $\frac{6}{3} = 2$ bowls of treats.

Thus, 6 dogs require 6 bowls of food, 3 bowls of water, and 2 bowls of treats, or $6 + 3 + 2 = 11$ bowls in total.

There are 11 bowls for every 6 dogs, and since there are $77 = 11 \times 7$ bowls, then there are 7 groups of 6 dogs, or $6 \times 7 = 42$ dogs.

18. Two standard six-sided dice are rolled. If the two numbers on the top faces are multiplied, which of the following products is most likely?
- (A) 4 (B) 6 (C) 9 (D) 15 (E) 8

Source: 2025 Gauss Grade 8 #18

Primary Topics: Counting and Probability

Secondary Topics: Probability | Counting | Operations

Answer: B

Solution:

18. In the grid shown, the first column and first row show the possible numbers that may appear on the top faces of the two dice.

The inside of the grid shows the product of the corresponding two numbers.

×	1	2	3	4	5	6
1	1	2	3	4	5	6
2	2	4	6	8	10	12
3	3	6	9	12	15	18
4	4	8	12	16	20	24
5	5	10	15	20	25	30
6	6	12	18	24	30	36

Of the results, a product of 4 appears 3 times, a product of 6 appears 4 times, a product of 9 appears 1 time, a product of 15 appears 2 times, and a product of 8 appears 2 times. Thus, of the possible products given, 6 is the most likely to occur.

19. How many ordered pairs of positive integers (m, n) are there so that $m^2 \times n = 2025$?

- (A) 3 (B) 7 (C) 4 (D) 5 (E) 6

Source: 2025 Gauss Grade 8 #19

Primary Topics: Number Sense

Secondary Topics: Operations | Exponents | Counting | Divisibility | Prime Numbers

Answer: E

Solution:

We begin by determining the prime factorization of 2025.

$$\begin{aligned} 2025 &= 25 \times 81 \\ &= 5 \times 5 \times 9 \times 9 \\ &= 5 \times 5 \times 3 \times 3 \times 3 \times 3 \\ &= 3^4 \times 5^2 \end{aligned}$$

There are exactly 15 positive factors of 2025. These are:

1, 3, 3^2 , 3^3 , 3^4 , 5, 5^2 , 3×5 , $3^2 \times 5$, $3^3 \times 5$, $3^4 \times 5$, 3×5^2 , $3^2 \times 5^2$, $3^3 \times 5^2$, and $3^4 \times 5^2$

We are asked to express 2025 as the product of two positive integers n and m^2 , where m^2 is a perfect square. Of the 15 positive factors, the following are perfect squares:

1, 3^2 , 3^4 , 5^2 , $3^2 \times 5^2$, and $3^4 \times 5^2$

These are all the possible values of m^2 , and so the values of m are: 1, 3, 5, 3^2 , 3×5 , and $3^2 \times 5$.

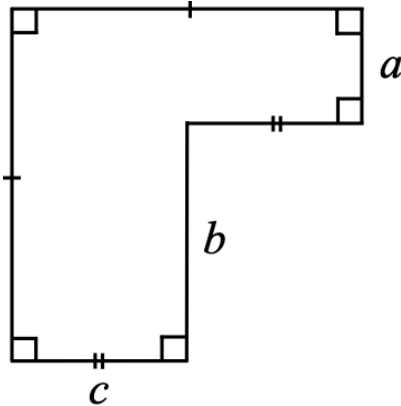
Thus, the ordered pairs of positive integers (m, n) for which $m^2 \times n = 2025$ are:

$(1, 3^4 \times 5^2)$, $(3, 3^2 \times 5^2)$, $(5, 3^4)$, $(3^2, 5^2)$, $(3 \times 5, 3^2)$, $(3^2 \times 5, 1)$

Therefore, there are 6 such ordered pairs.

It is interesting to note that each of the 6 values of n is also a perfect square. Can you see why this occurs?

20. In the diagram, each of a , b and c is greater than zero.



A 6-sided polygon, that alternates between horizontal and vertical sides, forms a shape like an upside down capital L. The perimeter of the polygon can be traced as follows: Starting at the top right vertex, move to the left some distance to the top left vertex and then down an equal distance to the bottom left vertex; then move c units to the right, then b units up, then c units to the right again, and then a units up to arrive back at the top right vertex.

Which of the following expressions is *not* equal to the perimeter of this polygon?

- (A) $4a + 4b$ (B) $a + b + 7c$ (C) $8c$ (D) $2a + 2b + 4c$ (E) $3a + 3b + 2c$

Source: 2025 Gauss Grade 8 #20

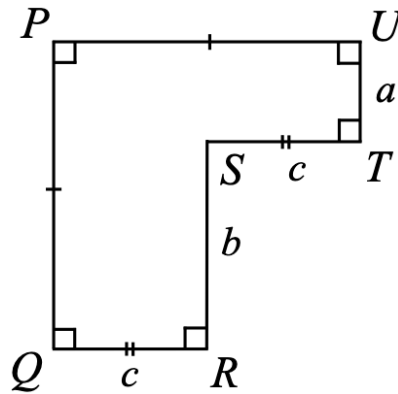
Primary Topics: Geometry and Measurement | Algebra and Equations

Secondary Topics: Prisms | Expressions | Measurement

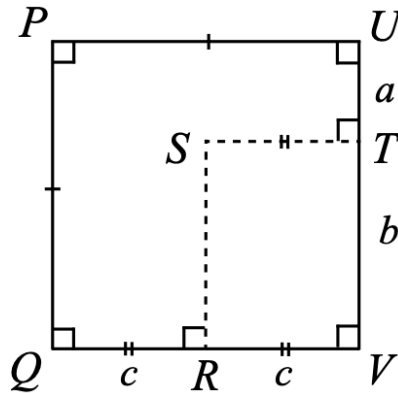
Answer: B

Solution:

In the first diagram shown, we label the vertices of the polygon and the length $ST = c$, since $ST = QR$.



Next, we extend UT by a length equal to SR , and we extend QR by a length equal to ST , as shown in the second diagram.



Each of the angles in the polygon is a right angle, and so these two extended line segments are perpendicular to each other and will meet at a point that we label V .

That is, $STVR$ is a rectangle with $TV = SR = b$ and $RV = ST = c$.

Each of the following expressions is equal to the perimeter of the original polygon

$$\begin{aligned}
 &PQ + QR + SR + ST + TU + PU \\
 &= PQ + QR + ST + SR + TU + PU \text{ (reordering the lengths)} \\
 &= PQ + QR + RV + TV + TU + PU \text{ (since } RV = ST \text{ and } TV = SR) \\
 &= PQ + QV + UV + PU \text{ (since } QR + RV = QV \text{ and } TV + TU = UV)
 \end{aligned}$$

which is the perimeter of $PQVU$.

Each of the angles in $PQVU$ is a right angle, and $PQ = PU$, and thus $PQVU$ is a square.

Since $PQ = UV = UT + TV = a + b$, and $PU = QV = QR + RV = c + c = 2c$, then $a + b = 2c$.

Summarizing, the perimeter of the original polygon is equal to the perimeter of square $PQVU$, and each side length of square $PQVU$ can be expressed as $a + b$ or as $2c$ since $a + b = 2c$.

If each of the 4 side lengths is expressed as $a + b$, then the perimeter of $PQVU$ (and thus the perimeter of the original polygon), is equal to $(a + b) + (a + b) + (a + b) + (a + b) = 4a + 4b$.

If 3 side lengths are expressed as $a + b$ and 1 side length is expressed as $2c$, then the perimeter is $(a + b) + (a + b) + (a + b) + (2c) = 3a + 3b + 2c$.

If 2 side lengths are expressed as $a + b$ and 2 side lengths are expressed as $2c$, then the perimeter is $(a + b) + (a + b) + (2c) + (2c) = 2a + 2b + 4c$.

If 1 side length is expressed as $a + b$ and 3 side lengths are expressed as $2c$, then the perimeter is $(a + b) + (2c) + (2c) + (2c) = a + b + 6c$.

Finally, if all 4 sides lengths are expressed as $2c$, the perimeter is $(2c) + (2c) + (2c) + (2c) = 8c$.

Of the expressions given, $a + b + 7c$ remains, and since $a + b + 7c = 2c + 7c = 9c$ is not equal to the perimeter, then the correct answer is (B).

21. In the diagram, each letter from A to H is equal to a different integer from 1 to 8.

A	E
B	F
C	G
D	H

Also,

- H is a perfect square and is 1 more than D
- 5 and 8 are in the same row
- C is a multiple of both G and D
- B is the largest prime number in the set
- The value of $B + G$ is even

What is the value of F ?

- (A) 2 (B) 6 (C) 1 (D) 7 (E) 8

Source: 2025 Gauss Grade 8 #21

Primary Topics: Number Sense | Other

Secondary Topics: Logic | Counting | Operations

Answer: A

Solution:

H is a perfect square and is 1 more than D .

The perfect squares from 1 to 8 inclusive are 1 and 4. Since H is one more than D , H cannot be equal to 1 (since $D \neq 0$), and so $H = 4$ and $D = 3$.

B is the largest prime number in the set, and so $B = 7$.

C is a multiple of both G and D . Since $D = 3$, then $C = 6$ and G is equal to either 1 or 2 (since 6 is a multiple of each of these).

The value of $B + G$ is even, and since $B = 7$, then $G = 1$.

The letters which have not been assigned values are A , E and F , and the integers which have not been assigned are 2, 5 and 8.

Since 5 and 8 are in the same row and A and E are the two remaining unassigned letters that are in the same row, then A and E are 5 and 8 in some order, which leaves $F = 2$.

22. $ABCD$ has vertices $A(-3, -2)$, $B(0, r)$, $C(6, 10)$, and $D(s, t)$. AB is parallel to CD , BC is parallel to AD , and $r < 0$. What is the value of $r + s + t$?
- (A) 10 (B) 11 (C) 12 (D) 13 (E) 14

Source: 2025 Gauss Grade 8 #22

Primary Topics: Geometry and Measurement

Secondary Topics: Coordinate Geometry | Polygons | Angles | Expressions

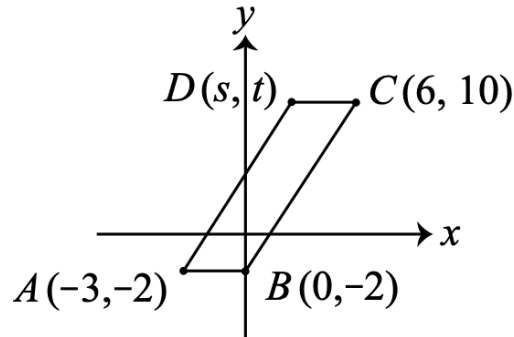
Answer: B

Solution:

Solution 1:

The opposite sides of $ABCD$ are parallel, and so $ABCD$ is a parallelogram. The opposite sides of a parallelogram are equal in length, and so $AB = CD$ and $AD = BC$.

Since the value of $r + s + t$ is equal to a constant, we may choose any location for $B(0, r)$ provided that it satisfies the given condition $r < 0$. We choose $r = -2$, so that the y -coordinate of $B(0, r)$ is equal to the y -coordinate of $A(-3, -2)$, and so AB is a horizontal line segment as shown.



In this case, the length of AB is equal to the positive difference between the x -coordinates of A and B , which is $0 - (-3) = 3$.

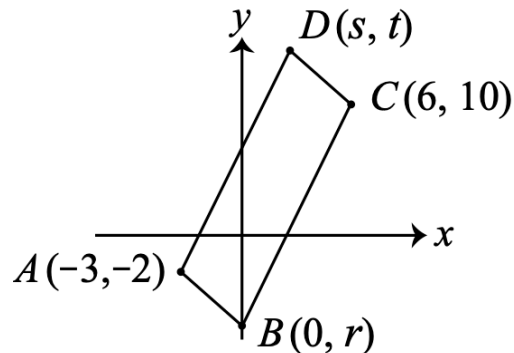
CD is parallel to AB , and so CD must also be a horizontal line segment. Thus, points $C(6, 10)$ and $D(s, t)$ have equal y -coordinates, and so $t = 10$. Further, CD has the same length as AB , and so $6 - s = 3$ or $s = 3$.

Therefore, the value of $r + s + t = -2 + 3 + 10 = 11$.

Solution 2:

The opposite sides of $ABCD$ are parallel, and so $ABCD$ is a parallelogram. The opposite sides of a parallelogram are equal in length, and so $AB = CD$ and $AD = BC$.

Since AB and CD are parallel and equal in length, then the vertical distance between A and B must equal the vertical distance between C and D , and the horizontal distance between A and B must equal the horizontal distance between C and D .



The vertical distance between two points is equal to the non-negative difference between their y -coordinates, and so $-2 - r = t - 10$ (assuming $r \leq -2$ and $t \geq 10$ as in the diagram).

Simplifying, we get $-2 + 10 = t + r$ and so $t + r = 8$.

The horizontal distance between two points is equal to the non-negative difference between their x -coordinates, and so $0 - (-3) = 6 - s$ (assuming $s < 6$ as in the diagram).

Simplifying, we get $0 + 3 = 6 - s$ or $s = 6 - 3$ and so $s = 3$.

Therefore, the value of $r + s + t = (r + t) + s = 8 + 3 = 11$. From Solution 1, we note that $r = -2$, $s = 3$, $t = 10$ are values satisfying the given conditions and for which $r + s + t = 11$.

23. The number 2013 is multiplied by a positive integer n . The last four digits of the result are 2025. What is the sum of the digits of the smallest possible value of n ?
- (A) 17 (B) 13 (C) 15 (D) 14 (E) 16

Source: 2025 Gauss Grade 8 #23

Primary Topics: Number Sense | Algebra and Equations

Secondary Topics: Operations | Equations Solving | Counting

Answer: E

Solution:

Solution 1:

Let the last three digits of n be abc . That is, n has units digit c , tens digit b and hundreds digit a . (By the end of this solution, we will have demonstrated why considering only the last three digits of n was sufficient.)

The units digit of the product $2013 \times n$ is equal to the units digit of $3 \times c$.

$$\begin{array}{r} 2013 \\ \times \quad abc \\ \hline \dots 2025 \end{array}$$

Since the units digit of the product $2013 \times n$ is 5, then the units digit of $3 \times c$ is 5, and so $c = 5$. (You should confirm for yourself that this is the only possible value of c .)

$$\begin{array}{r} 2013 \\ \times \quad 5 \\ \hline 10065 \end{array}$$

Continuing the long multiplication, the tens digit of n is b , and so the tens digit of $2013 \times n$ is equal to the units digit of $6 + 3b$, as shown.

Since the units digit of $6 + 3b$ is 2, then the units digit of $3b$ is 6, and so $b = 2$.

(You should confirm for yourself that this is the only possible value of b .)

$$\begin{array}{r} 2013 \\ \times \quad b5 \\ \hline 10065 \\ \dots 3b \\ \hline \dots 25 \end{array}$$

The multiplication completed to this point is shown to the right.

We have determined that the last two digits of the product $2013 \times n$ are 25 exactly when the last two digits of n are 25.

$$\begin{array}{r} 2013 \\ \times \quad 25 \\ \hline 10065 \\ 4026 \\ \hline \dots 25 \end{array}$$

Continuing the long multiplication, the hundreds digit of n is a , and so the hundreds digit of $2013 \times n$ is equal to the units digit of $1 + 0 + 2 + 3a$ (the 1 is the "carry" from the tens column).

Since the units digit of $3 + 3a$ is 0, then the units digit of $3a$ is 7, and so $a = 9$.

(You should confirm that this is the only possible value of a .)

$$\begin{array}{r} 2013 \\ \times \quad a25 \\ \hline 10065 \\ 4026 \\ \dots 3a \\ \hline \dots 025 \end{array}$$

The last three digits of $2013 \times n$ are 025 exactly when the last three digits of n are 925 (that is, $a = 9$, $b = 2$, $c = 5$ are the only possibilities for a , b , c).

The multiplication completed to this point is shown below.

$$\begin{array}{r}
 2013 \\
 \times 925 \\
 \hline
 10065 \\
 4026 \\
 18117 \\
 \hline
 1862025
 \end{array}$$

This shows that when $n = 925$, the last four digits of the product $2013 \times n$ are 2025, as required. Adding additional digits to n will increase the value of n , and since we are asked for the smallest possible value of n , we stop here.

Thus, the smallest possible value of n for which $2013 \times n$ has last four digits 2025, is $n = 925$, and so the sum of the digits of n is $9 + 2 + 5 = 16$.

Solution 2:

We begin by showing that every positive integer having last two digits 25 is a multiple of 25.

(It is worth noting that it is *not* true that every multiple of 25 has last two digits 25.)

All positive integers whose last two digits are 25, are 25 more than some non-negative multiple of 100.

That is, all positive integers whose last two digits are 25 can be expressed as $100k + 25$ for some integer $k \geq 0$.

Since $100k$ is divisible by 25, and 25 is divisible by 25, then $100k + 25$ is divisible by 25.

Thus, every positive integer whose last two digits are 25 is a multiple of 25, and so $2013 \times n$ is a multiple of 25.

Since $2013 = 3 \times 11 \times 61$ does not have a prime factor of 5, then $2013 \times n$ is a multiple of 25 exactly when n is a multiple of 25.

The last two digits of $2013 \times n$ are equal to the two-digit number formed by the last two digits of the product of 13 and the last two digits of n .

What are the last two digits of n ? Since n is a multiple of 25, then the last two digits of n could be 25, 50, 75, or 00.

(You should confirm that these are the only possibilities.)

Thus, the last two digits of $2013 \times n$ are equal to the last two digits of 13×25 or 13×50 or 13×75 or 13×00 , which are 25, 50, 75, and 00 respectively.

Since we require the last two digits of $2013 \times n$ to be 25, then the last two digits of n are 25.

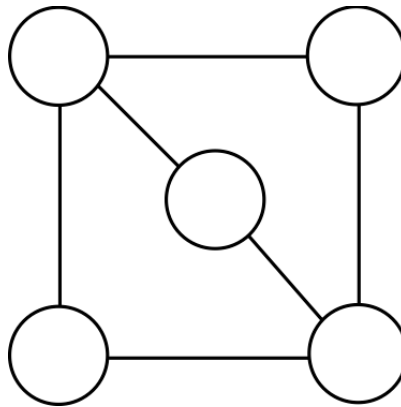
We have reduced the problem to finding the smallest value of the positive integer n with last two digits 25 so that the last four digits of $2013 \times n$ are 2025.

We substitute $n = 25, 125, 225, 325, 425, \dots$, and so on, in turn, into the product $2013 \times n$.

Evaluating these products, we determine that $2013 \times 925 = 1\,862\,025$ is the first time that the last four digits of $2013 \times n$ are 2025.

Thus, the smallest possible value of n for which $2013 \times n$ has last four digits 2025 is $n = 925$, and the sum of the digits of n is $9 + 2 + 5 = 16$.

24. In the diagram, circles are *connected* if they are joined by a line segment.



Each circle is filled with one integer so that

- the positive difference between each pair of integers in connected circles is d , and
- the sum of the five integers in the circles is 54.

For how many different values of d between 1 and 20 inclusive can the circles be filled in this way?

- (A) 4 (B) 12 (C) 8 (D) 20 (E) 16

Source: 2025 Gauss Grade 8 #24

Primary Topics: Other | Number Sense

Secondary Topics: Logic | Counting | Operations

Answer: E

Solution:

Suppose the integer in the centre circle is a .

Then each integer in a circle connected to the centre circle is either d more than a , which is $a + d$, or it is d less than a , which is $a - d$.

The integers in the two circles connected to the centre could both be $a + d$, as in Figure 1 below, or they could both be $a - d$, as in Figure 2, or one could be $a - d$ and one could be $a + d$, as in Figure 3.

We note that in the last case (Figure 3), swapping locations of the $a - d$ and the $a + d$ does not change the integers in the final two empty circles (since they still depend on $a + d$ and $a - d$), and thus does not change the sum of the five integers.

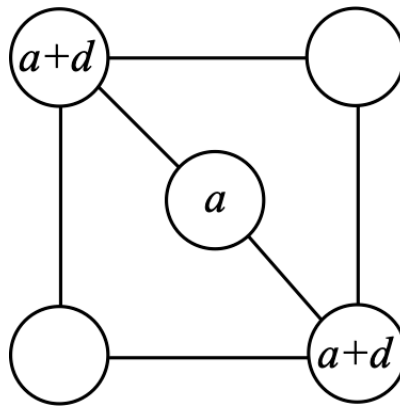


Figure 1

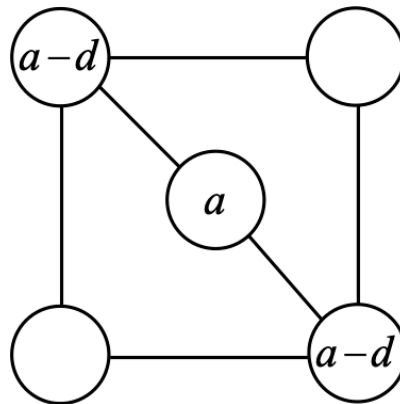


Figure 2

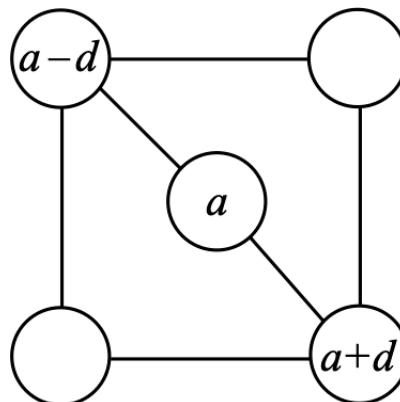


Figure 3

Next, we explain why it is possible to place integers into the remaining two circles (in each of the three cases above) so that the positive difference between each pair of integers in connected circles is d .

For the case that began in Figure 1, the integers in the two empty circles are either d more than $a + d$, which is $a + 2d$, or they are d less than $a + d$, which is a .

These integers could both be $a + 2d$, as in Figure 1a below, or they could both be a , as in Figure 1b, or one could be $a + 2d$ and one could be a , as in Figure 1c.

We note that in the last case (Figure 1c), swapping locations of the final two integers, $a + 2d$ and a , does not change the sum of the five integers.

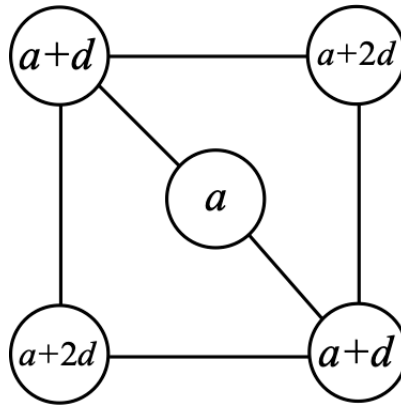


Figure 1a

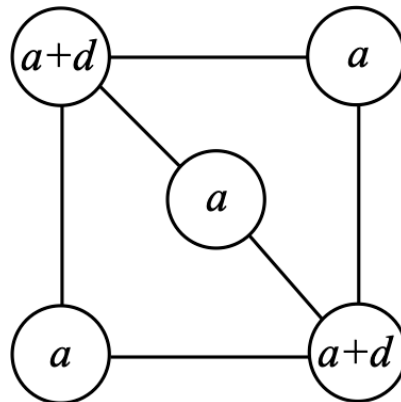


Figure 1b

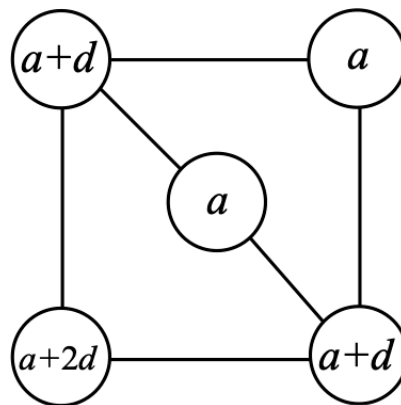


Figure 1c

For the case that began in Figure 2, the integers in the two empty circles are either d more than $a - d$, which is a , or they are d less than $a - d$, which is $a - 2d$.

These integers could both be a , as in Figure 2a below, or they could both be $a - 2d$, as in Figure 2b, or one could be a and one could be $a - 2d$, as in Figure 2c.

We again note that in the last case (Figure 2c), swapping locations of the final two integers, a and $a - 2d$, does not change the sum of the five integers.

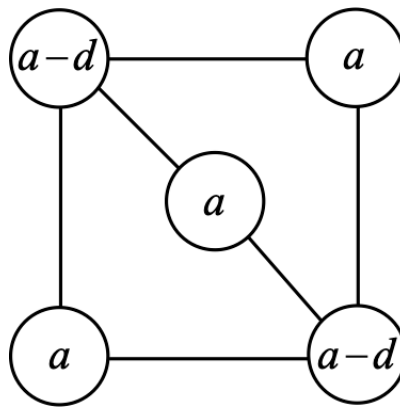


Figure 2a

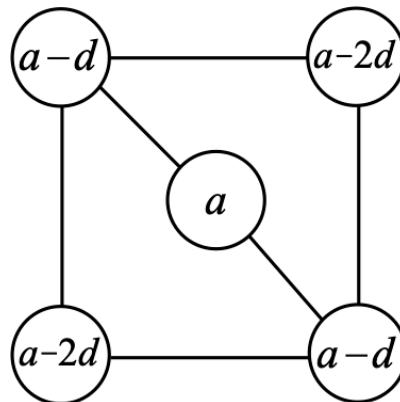


Figure 2b

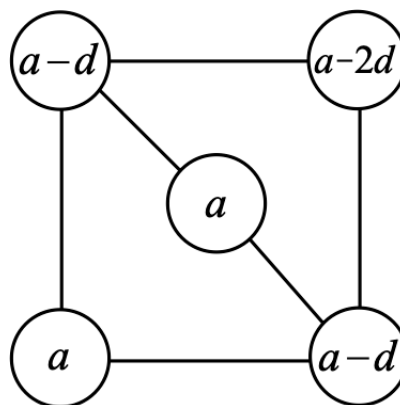


Figure 2c

Finally, for the case that began in Figure 3, each integer in an empty circle must have a positive difference of d with both $a - d$ and $a + d$.

The integer d more than $a - d$ is a , and the integer d less than $a - d$ is $a - 2d$.

The integer d more than $a + d$ is $a + 2d$, and the integer d less than $a + d$ is a .

Thus, a is the only integer that has a positive difference of d with both $a - d$ and $a + d$, and so the integers in the two empty circles must each be equal to a , as shown in Figure 3a.

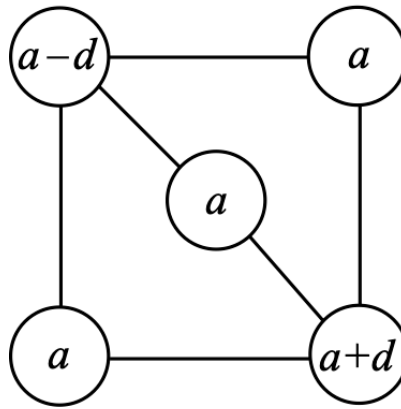


Figure 3a

Suppose that the sum of the five integers in the circles is S .

For Case 1a (which corresponds to Figure 1a), adding the five integers in the figure, we get $S = a + (a + d) + (a + 2d) + (a + d) + (a + 2d) = 5a + 6d$.

In the table below, we determine the value of S for each of the 7 cases.

Case 1a	Case 1b	Case 1c	Case 2a	Case 2b	Case 2c	Case 3a
$S = 5a + 6d$	$S = 5a + 2d$	$S = 5a + 4d$	$S = 5a - 2d$	$S = 5a - 6d$	$S = 5a - 4d$	$S = 5a$

We must determine the number of different integers d , between 1 and 20 inclusive, for which at least one of the seven expressions for S is equal to 54 and a is an integer.

Consider Case 1a, from which we get $5a + 6d = 54$.

Since both a and d are integers, and d is between 1 and 20 inclusive, we can systematically substitute values of d into this equation, and then solve for a to determine if a is an integer.

For example if $d = 1$, we get $5a + 6 \times 1 = 54$ or $5a = 48$.

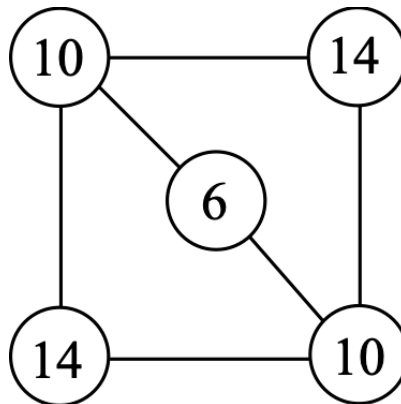
However, there is no integer a for which $5a = 48$ and so $d = 1$ is not a possible value of d in Case 1a.

Substituting $d = 2$ and $d = 3$ similarly give non-integer values of a .

When $d = 4$, we get $5a + 6 \times 4 = 54$ and so $5a = 30$ or $a = 6$.

In this case, the pair of integers $d = 4$ and $a = 6$ satisfy the equation $5a + 6d = 54$.

Substituting $d = 4$ and $a = 6$ into Figure 1a, we get the following diagram.



We can confirm that the positive difference between each pair of integers in connected circles is 4 (an integer between 1 and 20 inclusive), and the sum of the five integers in the circles is 54, as required.

Thus $d = 4$ is a possible value satisfying the given conditions.

We can systematically continue to substitute $d = 5, 6, 7, \dots, 20$ into $5a + 6d = 54$ and solve the equation to determine which values of d give integer values of a .

The next smallest value of d for which a is an integer is $d = 9$. In this case, we get $5a + 6 \times 9 = 54$ and so $5a = 0$ or $a = 0$.

We could continue in this systematic way, however since there are 20 possible values of d and 7 cases to check, this would take a while to complete. Instead, we might recognize that $d = 4$, $a = 6$ and

$d = 9, a = 0$ are both solutions to $5a + 6d = 54$.

Notice that from the first solution to the second, the value of d increases by 5, and the value of a decreases by 6.

Can you see why increasing d by 5 and decreasing a by 6 gives the next possible pair of integers for which $5a + 6d = 54$? (Hint: Take a close look at the left side of the equation.)

If we increase d by 5 again, and decrease a by 6, we get $d = 9 + 5 = 14$ and $a = 0 - 6 = -6$, and since $5a + 6d = 5 \times (-6) + 6 \times 14 = -30 + 84 = 54$, then $d = 14$ and $a = -6$ is a solution to the equation (and in fact, this is the next smallest value of d that works).

The final integer value of d between 1 and 20 inclusive for which $5a + 6d = 54$ is $d = 14 + 5 = 19$, and in this case $a = -6 - 6 = -12$ or $(d, a) = (19, -12)$

Therefore, Case 1a gives $d = 4, 9, 14$, and 19, or 4 values of d which satisfy the given conditions.

We continue in this way for each of the first four cases, and summarize all possible integer solutions for those cases in the table below.

Case 1a $5a + 6d = 54$ $(d, a) = (4, 6), (9, 0), (14, -6), (19, -12)$ $d = 4, 9, 14, 19$

Case 1b $5a + 2d = 54$ $(d, a) = (2, 10), (7, 8), (12, 6), (17, 4)$ $d = 2, 7, 12, 17$

Case 1c $5a + 4d = 54$ $(d, a) = (1, 10), (6, 6), (11, 2), (16, -2)$ $d = 1, 6, 11, 16$

Case 2a $5a - 2d = 54$ $(d, a) = (3, 12), (8, 14), (13, 16), (18, 18)$ $d = 3, 8, 13, 18$

Notice that after the first four cases shown above, all possible values of d from 1 to 20 inclusive satisfy the given conditions with the exception of $d = 5, 10, 15$, and 20.

In Case 3a we get, $5a = 54$ and so a is not an integer. Next, consider Case 2b, $5a - 6d = 54$.

Each value of d left to check ($d = 5, 10, 15, 20$) is a multiple of 5, and so $6d$ is a multiple of 5 for each of these possible values of d .

Since $5a$ is also a multiple of 5 for all possible integers a , then $5a - 6d$ is the difference between two multiples of 5, and thus is a multiple of 5.

However, the right side of the equation $5a - 6d = 54$ is not a multiple of 5 and so d cannot be equal to a multiple of 5.

In the final case, $5a - 4d = 54$, it is similarly not possible for d to be equal to a multiple of 5.

Thus d can be equal to each of the first 20 positive integers with the exception of 5, 10, 15, and 20, and so there are $20 - 4 = 16$ different possible values of d .

It is worth noting that there are many different ways to find the integer solutions to each of the 7 equations (cases) above. For example, the value of each of the terms $6d, 2d, 4d, -2d, -6d, -4d, 0d$ is even for all integers d , and the right side of each equation, 54, is also even. This means that in each equation, the value of $5a$ must be even, and so a is even. Further, when a is even, the units digit of $5a$ is 0. Since the units digit of 54 is 4, what do we now know about the units digit of each term containing a d , and in each case, what does that tell us about the possible values of d ?

25. The list 11, 12, 14, 23, 31, 44, 45, 46, 56, 64, 67, 74 can be arranged so that the units digit of each number matches the tens digit of the number that follows it. For example, 12, 23, 31, 11, 14, 44, 45, 56, 67, 74, 46, 64 is one such arrangement. How many such arrangements of the given list are possible?
- (A) 18 (B) 24 (C) 36 (D) 30 (E) 12

Source: 2025 Gauss Grade 8 #25

Primary Topics: Counting and Probability | Number Sense

Secondary Topics: Counting | Digits | Logic | Patterning/Sequences/Series

Answer: B

Solution:

We begin by recognizing that in the given list, each of the digits 1 through 7 occurs at least once as a units digit, and at least once as a tens digit.

For example, the digit 1 occurs twice as a units digit (11 and 31), and three times as a tens digit (11, 12 and 14).

Counting the number of times each of the digits 1 through 7 occurs as a units digit and as a tens digit, we get:

Digit	1	2	3	4	5	6	7
Number of times occurring as a units digit	2	1	1	4	1	2	1
Number of times occurring as a tens digit	3	1	1	3	1	2	1

The units digit of each number in the list matches the tens digit of the number that follows it.

This tells us that if we ignore the tens digit of the first number in the list and the units digit of the last number in the list, then the number of times that each digit occurs as a units digit must be equal to the number of times that it occurs as a tens digit.

Looking back to the table above, we see that this is true for all digits except 1 and 4.

Since the digit 1 occurs twice as a units digit and three times as a tens digit, then the tens digit of the first number in the list must be equal to 1.

Similarly, the digit 4 occurs four times as a units digit and three times as a tens digit, and so the units digit of the last number in the list must be equal to 4.

Ignoring the number 14 for a moment, we separate the 11 remaining numbers into two distinct lists, which we call A and B .

$$A : 11, 12, 23, 31 \quad B : 44, 45, 46, 56, 64, 67, 74$$

Each digit in A is less than or equal to 3, and each digit in B is greater than or equal to 4.

Since 14 is the only number given that does not appear in A or B , and 14 has a digit that appears in A and a digit that appears in B , then 14 is the only number that can TMconnectTM the numbers in A to those in B .

Further, this tells us that the numbers in A must be arranged and then placed before an arrangement of the numbers in B , with 14 appearing between the two arrangements.

Also, the arrangement of the numbers in A must begin and end with a 1, and the arrangement of the numbers in B must begin and end with a 4 (since 14 occurs between the two lists).

Next, we count the number of different ways to arrange the numbers in A , starting and ending with 1. We begin by recognizing that each of the digits 2 and 3 occurs exactly once as a units digit and once as a tens digit, and so 12, 23, 31 must appear together in this order (the two 2s must occur together and the two 3s must occur together).

The list must begin and end with a 1, and so there are 2 possible locations for the 11 and thus 2 possible arrangements of the numbers in A : 11, 12, 23, 31, and 12, 23, 31, 11.

Next, we count the number of different ways to arrange the numbers in B , starting and ending with 4. We begin by recognizing that each of the digits 5 and 7 occurs exactly once as a units digit and once as a tens digit, and so 45, 56 must appear together in this order (the two 5s must occur together), and 67, 74 must appear together in this order (the two 7s must occur together).

The arrangement ends with a 4, and thus cannot end with 45, 56, and so at least one more number must immediately follow 45, 56.

There are two such possibilities: 45, 56, 64, and 45, 56, 67, 74 (recall that 67, 74 must remain together), which leads to exactly two distinct cases to consider.

Case 1: 45, 56, 64 occur together in this order.

In this case, the remaining numbers are 44, 46, 67, and 74.

Since 44 has two equal digits, its location in the arrangement of the B list cannot change the first digit in the list (which must be 4), and cannot change the last digit in the list (which must also be 4), and thus we ignore 44 for the moment.

The remaining numbers, 46, 67, 74 must occur together in this order. Can you see why?

Since the blocks 45, 56, 64 and 46, 67, 74 must each occur together in their respective orders, this gives two possible arrangements of the B list (ignoring the 44).

These are: 45, 56, 64, 46, 67, 74, and 46, 67, 74, 45, 56, 64.

Next, we determine the number of different ways to place 44 into each of these arrangements.

In the 45, 56, 64, 46, 67, 74 arrangement, the 44 may appear at the start, at the end, or between the 64 and 46, which gives 3 different arrangements of the B list. (These are: 44, 45, 56, 64, 46, 67, 74, and 45, 56, 64, 46, 67, 74, 44, and 45, 56, 64, 44, 46, 67, 74.)

In the 46, 67, 74, 45, 56, 64 arrangement, the 44 may appear at the start, at the end, or between the 74 and 45, which gives 3 more arrangements of the B list, or 6 in total for Case 1.

Case 2: 45, 56, 67, 74 occur together in this order.

In this case, the remaining numbers are 44, 46, and 64.

We again begin by ignoring 44 for the moment.

The remaining numbers, 46, 64 must occur together in this order.

Since the blocks 45, 56, 67, 74 and 46, 64 must each occur together in their respective orders, this gives two possible arrangements of the B list (ignoring the 44).

These are: 45, 56, 67, 74, 46, 64 and 46, 64, 45, 56, 67, 74.

In the 45, 56, 67, 74, 46, 64 arrangement, the 44 may appear at the start, at the end, or between the 74 and 46, which gives 3 more different arrangements of the B list.

In the 46, 64, 45, 56, 67, 74 arrangement, the 44 may appear at the start, at the end, or between the 64 and 45, which gives 3 more arrangements of the B list, or 6 in total for Case 2.

Thus, there are a total of $6 + 6 = 12$ different ways to arrange the B list.

There are 2 different ways to arrange the numbers in list A , 12 different ways to arrange the numbers in list B , and exactly 1 way to place the number 14 between arrangements of each of the two lists.

Thus, the total number of arrangements of the given list is $2 \times 12 \times 1 = 24$.
